

Sharing Collateral Data and Small Firm Credit Access

1. Motivation

Collateral is central to credit access because it gives lenders a claim they can seize if the borrower defaults, thereby relaxing the limits that asymmetric information and weak contract enforcement place on lending. For small firms, however, the assets that could play this role are difficult to verify and secure, especially in emerging markets. Lenders cannot verify whether a small firm asset is already pledged, and to whom, and even when they can, they cannot enforce their claim on it. The result is that small firms hold productive assets they cannot borrow against. To deal with this, many countries have implemented collateral registries. Whether these registries reach the underserved firms they target, and how they should be designed, remain open empirical questions.

Registries can raise credit access through two main channels. First, a pledgeability channel through which registered claims that are verifiable and enforceable make existing assets usable collateral, expanding credit access. Second, a disclosure channel by which public registry reveals to every rival lender that a borrower holds assets that have passed a lender's screening, converting a private lending relationship into a market-wide signal. Disclosure allows outside lenders to compete for borrowers who were previously captive, which lowers their cost of credit and expands credit access. However, it also erodes the rents that relationship lenders earn on those borrowers, rents that may fund lending to the smallest and most opaque firms that have nothing to register.

This proposal quantifies the effect of opening collateral information on small firm credit access. We study the Peru's 2005 open collateral registry policy and use administrative credit registry data to identify how this policy reshaped small business finance and lender competition.

2. Research Question and Contribution

We ask how easing barriers to collateral information sharing reshapes lender competition and small business outcomes, and whose information lenders choose to share. The project separates two channels: the pledgeability channel, under which collateral becomes enforceable and secured credit expands (Campello and Larrain, 2016; Calomiris, Larrain, Liberti, and Sturgess, 2017); and the disclosure channel, under which collateral becomes visible and incumbents' informational rents erode, therefore borrowers can be contested (Rajan, 1992; Petersen and Rajan, 1995).

Our proposal makes three main contributions. First, the paper will provide the first causal evidence on how collateral information disclosure shapes lender competition, distinguishing the disclosure channel from pledgeability. Second, it bridges the information-sharing literature with the collateral-registry literature by treating registration as an endogenous, lender-borrower-specific disclosure decision, and by studying how registries can reshape informational rents. Third, it speaks to the open banking debate. Existing regimes share information about a borrower's behavior (e.g., transactions, loan repayment), while none shares information about assets and legal claims; neither do which lender already knows about which borrower has collateral.

3. Setting

Peru created the Sistema Informativo de Garantías Mobiliarias (SIGM) in 2005, a national electronic registry of movable assets. This policy has three important features for testing the channels through which collateral impacts lending.

First, registration is the creditor's choice, not a mandate. Filing an electronic notice confers priority and enforceability against third parties but publishes the borrower's identity and the collateral description to a publicly searchable database. Registration therefore trades formal seniority for the potential erosion of informational rents. Second, enforcement arrived in stages. The registry opened nationally on March 3, 2025, while the Judiciary connected to the SIGM court by court. Since seniority is only worth when a court can execute, the value of a registered guarantee increased at different dates in different judicial districts. Third, the policy only targeted movable collateral, widening the menu of pledgeable assets. Thus, borrowers secured on real estate constitute a placebo group, while borrowers with movable guarantees, or none, are the ones benefited by the policy.

Peru's credit market offers a key additional advantage to our paper. It exhibits a sharp segmentation in informational technology across lender types. Commercial banks lend on hard information such as financial statements, registrable collateral, and credit scores, while a dense microfinance sector lends on soft, relationship-based information to small and micro firms. Thus, we can provide novel insights on how open collateral data reshapes lender competition in markets with lender specialization in information acquisition.

4. Data

We use Peru's Reporte Crediticio Consolidado (RCC), the universe of outstanding debt for every borrower-lender pair, with lender-reported guarantee records by class (movable, immovable, public programs). We combine this dataset with loan origination data for the small and micro segment, including interest rates, lender type, and branch locations.

The RCC records whether a relationship carries a guarantee and of what class; it does not record SIGM filings. The tests described in this proposal require filing-level SIGM data (date, creditor, debtor identifier, asset class, etc.) together with the effective date on which each Superior Court gained access to the SIGM module. Matched to the RCC by borrower and lender identifiers, these data allow our preliminary evidence reported in Section 6 to be decomposed into registered and unregistered movable collateral.

5. Conceptual Framework and Testable Predictions

We organize the analysis around three building blocks; (i) lenders' incentives to share collateral information, (ii) the competitive response of rival lenders who observe that disclosure, and (iii) the resulting borrower-level outcomes in credit access, pricing, and real activity.

In the first part of the paper, we will model the registration decision. Registering a movable guarantee confers seniority and enforceable priority against third-party claims. Then, the value of registration increases with the liquidation value of the collateral, with competition threats summarized by the probability of borrowers to have other lenders, and with local enforcement quality. The staggered Court rollout identifies the enforcement value of registration, while the registration decision itself is identified from the national opening interacted with lender type and local competition.

Lenders weigh the benefits of seniority against the cost of disclosure, which erodes informational rents and is larger for relationship lenders whose rents derive from private knowledge of borrower quality. This cost scales up with local competition because registration is only dangerous when rival lenders can observe the borrower's information. We test the following predictions:

1. Conditional on borrower size and collateral type, commercial banks register movable guarantees at higher rates than microfinance institutions.

2. Registration rises after a borrower's judicial district connects to the SGM, since seniority is more valuable where it can be executed.
3. For relationship lenders, registration is decreasing, or less increasing, in local rival presence; for hard-information banks the effect is attenuated or reversed.
4. Relationship lenders concentrate registration on low-rent borrowers (larger, multi-lender, formal firms) and are less likely to register high-rent, captive borrowers.

In the second part, we will examine the competitive response. A registered notice reveals a screened borrower with enforceable collateral, and a rival lender can make a competing offer. Poaching intensity rises with the number of rival branches in the borrower's city, and the incumbent's hold-up rents erode as competition intensifies (Rajan, 1992; Petersen and Rajan, 1995).

Within relationship-lender portfolios, the rents on collateralized borrowers may help fund the costly relationship investment in the smallest, most opaque clients. When registration and poaching strip those rents, the lender's participation constraint on the opaque segment tightens. We will test the following predictions:

1. Poaching concentrates on borrowers with registered movable guarantees and increases with rival branch density in the borrower's city.
2. Interest rates fall for registered, contested borrowers, and spreads compress most where rival presence is dense.
3. Within relationship-lender portfolios, rates rise or credit contracts for the smallest borrowers with nothing to register, more so at lenders that lost more to poaching.

In the third part of the paper, we will quantify the net effect on small business finance as the balance of two reallocation forces. Contested borrowers receive cheaper, larger credit from new lenders, potentially reducing capital misallocation; squeezed opaque borrowers lose as relationship lenders tighten participation constraints. Marginal productivity is proxied by value added per unit of capital. We test the following predictions:

1. Total credit and average rates improve for the registered, contestable segment, particularly for higher-return firms, implying a decline in capital misallocation.
2. Credit access deteriorates at the unsecured, opaque bottom of relationship lenders' portfolios.

6. Identification and Preliminary Evidence

We exploit several sources of variation; time (the registry opens in March 2025), collateral type (movable versus real-estate guarantees, fixed before the reform), enforcement (staggered court connection across judicial districts), local competition (pre-reform rival branch density in the borrower's district), lender type (banks versus relationship lenders), and, once SGM filings are matched, the registration decision itself.

Among small borrowers with a recorded guarantee between July 2024 and January 2025, those whose guarantee is only movable increase business debt, relative to borrowers with real-estate collateral in the same district, industry, size bin, and quarter, by about 12 percent by October 2025. The pre-reform coefficients are flat, and consumer and mortgage debt do not move. How this credit expansion operates, which lenders expand, and whether the registered firms drive it, is exactly what filing-level data will show.

References

- Calomiris, C. W., M. Larrain, J. Liberti, and J. Sturgess (2017). How collateral laws shape lending and sectoral activity. *Journal of Financial Economics* 123(1), 163–188.
- Campello, M. and M. Larrain (2016). Enlarging the contracting space: collateral menus, access to credit, and economic activity. *Review of Financial Studies* 29(2), 349–383.
- Hauswald, R. and R. Marquez (2006). Competition and strategic information acquisition in credit markets. *Review of Financial Studies* 19(3), 967–1000.
- He, Z., J. Huang, and J. Zhou (2023). Open banking: credit market competition when borrowers own the data. *Journal of Financial Economics* 147(2), 449–474.
- Padilla, A. J. and M. Pagano (1997). Endogenous communication among lenders and entrepreneurial incentives. *Review of Financial Studies* 10(1), 205–236.
- Pagano, M. and T. Jappelli (1993). Information sharing in credit markets. *Journal of Finance* 48(5), 1693–1718.
- Petersen, M. A. and R. G. Rajan (1995). The effect of credit market competition on lending relationships. *Quarterly Journal of Economics* 110(2), 407–443.
- Ponticelli, J. and L. S. Alencar (2016). Court enforcement, bank loans, and firm investment: evidence from a bankruptcy reform in Brazil. *Quarterly Journal of Economics* 131(3), 1365–1413.
- Rajan, R. G. (1992). Insiders and outsiders: the choice between informed and arm’s-length debt. *Journal of Finance* 47(4), 1367–1400.
- Yu, T. (working paper). Data as collateral: open banking for small business lending. Year and venue to be confirmed.